1. Let $f_n$ denote the number of ways of tossing a coin $n$ times such that successive heads never appear. Argue that

$$f_n = f_{n-1} + f_{n-2} \quad n \geq 2, \text{where } f_0 = 1, f_1 = 2$$

*Hint:* How many outcomes are there that start with a head, and how many start with a tail? If $P_n$ denotes the probability that successive heads never appear when a coin is tossed $n$ times, find $P_n$ (in terms of $f_n$) when all possible outcomes of the $n$ tosses are assumed equally likely. Compute $P_{10}$.

2. An urn contains $n$ red and $m$ blue balls. They are withdrawn one at a time until a total of $r, r \leq n$, red balls have been withdrawn. Find the probability that a total of $k$ balls are withdrawn.

3. A high school student is anxiously waiting to receive mail telling her whether she has been accepted to a certain college. She estimates that the conditional probabilities of receiving notification on each day of next week, given that she is accepted and that she is rejected, are as follows:

| Day | $P(\text{mail} \mid \text{accepted})$ | $P(\text{mail} \mid \text{rejected})$ |
|---|---|---|
| Monday | .15 | .05 |
| Tuesday | .20 | .10 |
| Wednesday | .25 | .10 |
| Thursday | .15 | .15 |
| Friday | .10 | .20 |

She estimates that her probability of being accepted is .6.

(a) What is the probability that she receives mail on Monday?

(b) What is the conditional probability that she received mail on Tuesday given that she does not receive mail on Monday?

(c) If there is no mail through Wednesday, what is the conditional probability that she will be accepted?

(d) What is the conditional probability that she will be accepted if mail comes on Thursday?

(e) What is the conditional probability that she will be accepted if no mail arrives that week?

4. In a certain contest, the players are of equal skill and the probability is $\frac{1}{2}$ that a specified one of the two contestants will be the victor. In a group of $2^n$ players, the players are paired off against each other at random. The $2^{n-1}$ winners are again paired off randomly, and so on, until a single winner remains. Consider two specified contestants, $A$ and $B$, and define the events $A_i, i \leq n, E$ by

$A_i$: $A$ plays in exactly $i$ contests:
$E$: $A$ and $B$ play each other.

(a) Find $P(A_i), i = 1, \ldots, n$.

(b) Find $P(E)$.

(c) Let $P_n = P(E)$. Show that

$$P_n = \frac{1}{2^n - 1} + \frac{2^n - 2}{2^n - 1} \left(\frac{1}{2}\right)^2 P_{n-1}$$

and use this formula to check the answer you obtained in part (b).

5. Suppose that $n$ independent tosses of a coin having probability $p$ of coming up heads are made. Show that the probability that an even number of heads results is $\frac{1}{2}[1 + (1 - 2p)^n]$.

6. For a nonnegative integer-valued random variable $N$, show that

(a)

$$E[N] = \sum_{i=1}^{\infty} P\{N \geq i\}$$

(b)

$$E[N^2] = \sum_{i=1}^{\infty} (2i - 1) P(N \geq i)$$